Huy Nguyen

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Research Summary

Ph.D. researcher specializing in the systems-level architecture and distributed acceleration of Large Language Models (LLMs). My work focuses on pushing the physical limits of hardware utilization, from bare-metal CUDA/Cutlass kernel engineering, to designing topology-aware 5D parallel training engines. I build the underlying computational and cross-GPU interconnect infrastructure required to eliminate network bubbles and compute bottlenecks at massive scale.

Domain Expertise & Technologies

Languages & Frameworks: C/C++, Python, CUDA, Triton, PTX, PyTorch, Cutlass CuTe, CuTeDSL, NCCL, MPI

GPU Programming & Architecture: Asynchronous Pipelining, Persistent Kernels, Warp Specialization, Memory Coalescing, NVIDIA Hopper (TMA, WGMMA), NVIDIA Ampere

Distributed Systems: 5D Parallelism (DP, TP, PP, CP, EP), Customizing Collective Optimization (`all_reduce`, `all_to_all`, ...)

Education

University of Oregon *Sept 2024 – Present*
PhD in Computer Science

- GPA: 4.16/4.0, Excellent Degree
- Research Topic: Efficient LLMs
- Supervisor: Assoc. Prof. Thien Huu Nguyen, thienn@uoregon.edu

Hanoi University of Science and Technology *Sept 2018 – May 2023*
BS in Computer Science

- GPA: 3.71/4.0, Excellent Degree
- Advisor: Dr. Linh Ngo Van, linhnv@soict.hust.edu.vn

Experience

Amazon Web Services (AWS), Annapurna Labs *Applied Scientist Intern* *Cupertino, CA, USA Jun 2026 – Sep 2026*

- Accelerated parallel speculative decoding for Large Language Model (LLM) inference.

Microsoft Research *Research Intern* *Redmond, WA, USA Jun 2025 – Sep 2025*

- Engineered and optimized "Megakernels" on NVIDIA GPUs to accelerate Large Language Model (LLM) inference and reduce memory-bound latency.
- Designed low-level scheduling methods and custom fused operators using CUDA/Triton, maximizing hardware utilization and minimizing kernel launch overhead.

VinAI Research *AI Resident* *Hanoi, Vietnam Jul 2022 – Jul 2024*

- Supervisors:
  - Asst. Prof. Luu Anh Tuan, anhtuan.luu@ntu.edu.sg
  - Assoc. Prof. Thien Huu Nguyen, thienn@uoregon.edu
- Projects:
  - **A Spectral Viewpoint on Continual Relation Extraction**

- **Transitioning Representations between Languages for Cross-lingual Event Detection via Langevin Dynamics**

Open Source Contributions

- **Unsloth (PR #238)**: Engineered a highly optimized rotary embedding (RoPE) kernel using OpenAI Triton.
  - Redesigned memory access patterns to significantly improve throughput for LLM inference and training.

Projects

Custom 5D Parallel LLM Training Engine *PyTorch, NCCL, Distributed Systems*

- Engineered a Megatron-style distributed LLM trainer from scratch, implementing Data (DP), Tensor (TP), and Pipeline Parallelism (PP) with micro-batch scheduling and zero-bubble gradient routing.
- Architected a topology-aware Context Parallelism (CP) backend, combining DeepSpeed Ulysses for intra-node NVLink saturation and Zigzag Ring Attention for inter-node scaling.
- Implemented Expert Parallelism (EP) for Mixture-of-Experts (MoE) architectures, utilizing custom autograd wrappers and `all_to_all` collectives for token dispatch and dynamic capacity-factor load balancing.

High-Performance GPU Kernel Suite *CUDA, Triton, Cutlass CuTe*

- Engineered custom GEMM and FlashAttention kernels from scratch using plain CUDA and Cutlass CuTe, progressively optimizing memory access, coalescing, and multi-level tiling to outperform cuBLAS.
- Utilized advanced NVIDIA Hopper hardware features, including Tensor Memory Accelerator (TMA), WGMMA instructions, asynchronous pipelining, and warp specialization.

Vistral-7B-Chat *Vistral*

- Introduced Vistral-7B-chat, a SOTA conversational Large Language Model for Vietnamese. Outperforms ChatGPT and Gemini in Vietnamese benchmarks. Achieved over 100,000 downloads.

Publications

CRISP: Cliff-awaRe Input-adaptive Sparse Prefilling with Structural-Mass-Motivated Routing EMNLP 2026
Huy Huu Nguyen, Chien Van Nguyen, Franck Dernoncourt, Ryan A. Rossi, Linh Ngo Van, Jieyang Chen, Thien Huu Nguyen

Lizard: An Efficient Linearization Framework for Large Language Models ACL 2026
Chien Van Nguyen*, ***Huy Huu Nguyen****, Ruiyi Zhang, Hanieh Deilamsalehy, Puneet Mathur, Viet Dac Lai, Haoliang Wang, Jayakumar Subramanian, Ryan A. Rossi, Trung Bui, Nikos Vlassis, Franck Dernoncourt, Thien Huu Nguyen

Towards Fast and Accurate Modeling for Cross-Lingual Label Projection ACL 2026
Thang Le, ***Huy Huu Nguyen***, Anh Tuan Luu, Thamar Solorio, Thien Huu Nguyen

Massively Multilingual Instruction-Following Information Extraction ACL 2025 Findings
Thang Le, ***Huy Huu Nguyen***, Luu Anh Tuan, Thien Huu Nguyen

Taipan: Efficient and Expressive State Space Language Models with Selective Attention Arxiv
Chien Van Nguyen, ***Huy Huu Nguyen***, Trung Bui, Viet Dac Lai, Franck Dernoncourt, Thien Huu Nguyen

A Spectral Viewpoint on Continual Relation Extraction EMNLP 2023 Findings
Huy Nguyen, Chien Nguyen, Linh Ngo, Anh Luu, Thien Nguyen

Transitioning Representations between Languages for Cross-lingual Event Detection via Langevin Dynamics EMNLP 2023 Findings
Chien Nguyen, ***Huy Nguyen***, Franck Dernoncourt, Thien Nguyen

Academic Service

Conference Reviewer: ACL ARR 2023, ACL ARR 2024, ACL ARR 2025

Awards

Prestigious Recognition Academic Excellence

- Lokey Award (2024-25)
  - Prestigious award from the Department of Computer Science, University of Oregon.
  - Granted to top graduate students in science fields.
  - Funded by Mr. Lorry I. Lokey, allocated by UO Division of Graduate Studies.

First Place Award AI Competitions

- BKAI-NAVER Challenge 2022 (May 2022)
  - Task: Intent Detection and Slot Filling for smarthome systems.